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REVIEW

Drug-induced oral lichenoid reactions: a real clinical entity? A systematic review

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Abstract
Purpose Drug-induced oral lichenoid reactions (DIOLRs) have been extensively reported in the literature, but the validity of the causality relationship between any drug and the oral lichenoid lesions (OLLs) still remains questionable. We sought to determine whether this causality relationship really exists, whether a resolution of the oral lesions upon withdrawal occurs, and what the most common alleged offending medications are.
Methods Nine electronic databases from January 1966 to December 2016 were systematically searched to identify all relevant studies selected with specific inclusion criteria (a clinical and histopathological diagnosis of DIOLRs, and clearly statement on the systemic offending medication). Searched terms included but not limited to oral lichen planus/oral lichenoid lesions/oral lichenoid reactions, the adverse effects of medication, and drug-induced. Statistical analyses conducted.
Results The search retrieved a total of 817 articles, of which only 46 were included into a qualitative synthesis: 40 case reports/series and 6 studies. The causality assessment was done only in 14.8% of cases with the C-D-R protocol. The Naranjo algorithm was not reported in the majority of cases (98.2%). Culprit medication was withdrawn in 68.5% of the cases, obtaining a partial or complete resolution without treatment in 16.7% of cases and with treatment in 27.7% of cases. The median number of culprit medication(s) described was 1 with the most frequent ones being Methyldopa (20.37%), Interferon (IFN)-alpha (11.11%), and Imatinib and Infliximab (9.26%).
Conclusion This systematic review demonstrated that there is no strong scientific evidence to support the causal relationship between any drug and oral lichenoid lesions; therefore, in all reviewed cases, we must question whether the DIOLRs represent a real and separate clinical entity. Further and more thorough investigations using one of the available algorithms for adverse drug reaction are warranted.

Keywords Oral lichen planus · Oral lichenoid lesions · Oral licheniod reactions · Diagnosis · Management · Medication · Drug

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Introduction

In 2000, the definition of an adverse drug reaction described by the WHO [1] was modified by Edwards and Aarons [2] as follows: "an appreciably harmful or unpleasant reaction, resulting from an intervention related to the use of a medicinal product, which predicts hazard from future administration and warrants prevention or specific treatment, or alteration of the dosage regimen, or withdrawal of the product." They

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suggested that the terms “adverse reaction” and “adverse effect” can be used interchangeably, with the understanding that adverse reaction refers to the patient standpoint, whereas an adverse effect refers to the drug standpoint. The authors also distinguished an "adverse effect" and "adverse reaction" from an “adverse event,” indicating with the first an adverse outcome caused by some action of a drug, and with the latter an adverse outcome manifesting when a patient is on a drug, and it is not or not necessarily caused by that drug.

Although less common than in the skin, adverse reactions of the oral cavity may occur [3], and may present with a wide variety of clinical manifestations, such as hyposalivation/xerostomia, dysgeusia, oral burning, dyskinesia, apthous and non-aphtous like ulcer, oral lichenoid lesions (OLLs), bullous disorders (EM, SJS, PV, SLE), pigmentations, fibrovascular and keratosis/epithelial hyperplasia, dysesthesia, osteonecrosis of the jaws, infections, angioedema, and malignancies [3–7], or hypersalivation as in the neuroleptic malignant syndrome (NMS), an idiosyncratic neurological condition with potentially lethal consequences associated with the first and second generation antipsychotic use [8].

Of course, all those oral adverse reactions are strictly depended on patient’s factors, such as age, gender, genetic background in absorption, metabolism, and disposition, susceptibility of the patient, as well as medication factors such as category of medications, dosage, timing, route of administration, pharmacokinetic or pharmacodynamics anomalies, interactions with other drugs [6, 7].

The pathogenetic mechanism seems to be very complex and include two different types of reaction: type A, which are dose-dependent and based on the pharmacology of the drug, and type B, immunologic (antibody or T-cell mediated) and non-immunologic, such as direct mucosal/submucosal toxicity as seen in oral ulcer induced by nicorandil [9] or by conventional and targeted anticancer therapy [10].

Lichenoid drug eruption (LDE) as a variant of lichen planus (LP) was first described in 1929 with the first cases of oral involvement described in 1949 on military personnel in the South-West Pacific, treated prophylactically with mepacrine, an antimalarial drug [11]. Since 1949, many cases of drug-induced oral lichenoid reactions (DIOLRs) have been reported in the literature [7, 12, 13] and recently grouped under the category of oral lichenoid lesions (OLLs) [14, 15].

Although they clinically and histopathologically appear similar to oral lichen planus [9], several differences have been described. Clinically, DIOLRs seem to have a predominant erosive component [16], with a unilateral distribution and a positive basal cell cytoplasmic (BCC) antibody reaction [17], and mainly affect the buccal mucosa followed by lips and tongue [18].

Histologically, DIOLRs seem to have a diffuse inflammatory infiltrate consisting not by lymphocytes predominantly, but mixed with plasma cells and eosinophils, located deeper in the lamina propria [12, 18]. Additionally, those lesions may show parakeratosis, presence of colloid bodies in the epithelium, exocytosis into the upper epidermal layer, and a perivascular inflammatory cell infiltrate: features more common in discoid lupus erythematous (DLE) than in OLP [12, 18, 19].

Although DIOLRs have been extensively documented in the literature, in this systematic review, we sought to determine the validity of the causality relationship via the use of one of two (or both) most common protocols for adverse drug reactions: challenge–dechallenge–rechallenge (C-D-R) protocol and/or the Adverse Drug Reactions Probability Scale (APS), also known as Naranjo algorithm [20]. However, different algorithms, such as the WHO–UMC system [21], Gardner method, Jones algorithm, or Kramer questionnaire, currently validated in the scientific literature to evaluate ADRs [22], were also considered if used in the included papers. Additionally, we evaluated the complete or partial resolution of the oral lesions upon withdrawal without treatment and the most common alleged offending medication(s).

As a secondary endpoint, we sought to evaluate the clinical and histopathological features of DIOLRs.

Materials and methods

Selection strategy

The present study followed the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines [23], illustrating the outcomes of the literature searches and article selection process (Fig. 1). We performed a computerized literature search for articles published in nine databases from January 1966 to December 2016: Ovid MEDLINE, Embase, CINAHL, Scopus, Web of Science, Cochrane Libraries, ProQuest, ClinicalTrials.gov, and Google Scholar. The searches were concentrated on the two main topics of oral lichen planus/oral lichenoid lesions/oral lichenoid reactions and the adverse effects of medication, and drug-induced. We searched these concepts using controlled vocabulary particular to each database and keywords. MeSH terms included, but not limited to, Lichen Planus, Oral, Lichenoid lesions, Lichenoid reactions, Lichenoid eruptions AND Mouth, and Drug Related Side Effects and Adverse Reactions. The Ovid MEDLINE search was altered for each database. Citations and abstracts of retrieved articles were exported into RefWorks, and duplicates were removed. The complete search strategies for the three main database Ovid MEDLINE, Embase, and Scopus are included in the Appendix.

In addition, all references retrieved from all included articles were further examined manually to identify other relevant articles that were not detected by the initial search.

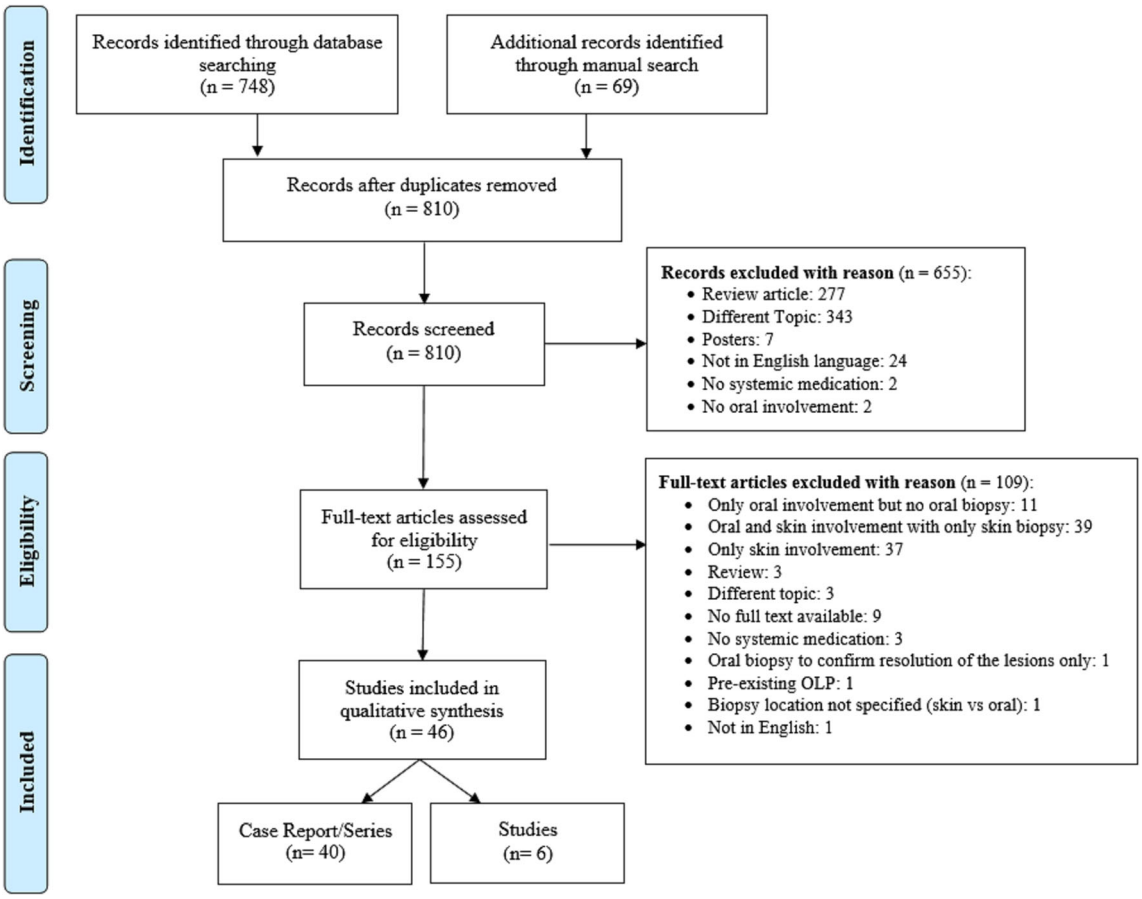

Fig. 1 PRISMA flow diagram of process of identification and screening of the included articles

Selection criteria

After eliminating duplicates, two investigators (GF and JHS) independently evaluated titles, abstract and full text of all the included articles. Any disagreement was discussed and resolved by consensus between the two investigators.

The systematic review included the following: (i) articles (case reports, case series, cohort studies, and case-control) published between January 1966 and December 2016 in the English language, (ii) cases with clinical features of DIOLRs with or without cutaneous and/or other mucous membranes involvement, (iii) only cases where diagnosis of DIOLRs was based on a histopathological examination (hematoxylin and eosin stain (H&E)) via an oral biopsy with or without a direct immunofluorescence (DIF) analysis, (iv) cases where the culprit medication(s) was/were clearly stated, and (v) cases in which oral lesions appeared during or after administration of systemic medications only.

We excluded randomized and non-randomized trials because they are interventional in nature and cross-sectional studies as they are performed at a single point in time or time interval, so unsuitable to establish a possible causative relationship. We also excluded case(s) with (i) oral involvement but no oral biopsy was taken, (ii) oral and cutaneous involvement with only a cutaneous biopsy or where the location of the biopsy was not clearly specified (skin viz. oral), (iii) only cutaneous lesions, (iv) oral lichenoid reactions induced by dental materials, and (v) preexisting OLP/OLLs.

Data extraction

Studies that met the inclusion criteria were independently reviewed by two investigators (GF and JS) who extracted and tabulated in an Excel sheet the following information: year of publication, country of publication, age, gender, number of patients (only for case series), culprit medication, personal history of allergy, family history of allergy, past medical history (PMH), concurrent medications at the time of DIOLRs onset, latency period (defined as the interval between the intake of the suspected offending medication and the development of oral lesions), clinical features of oral lesions, presence of cutaneous and/or other mucosal lesions, histopathological features with H&E, direct immunofluorescence (DIF) analyses, withdrawal of the alleged offending medication, partial resolution upon withdrawal without treatment, complete resolution upon withdrawal without treatment, type of treatment, partial resolution upon

withdrawal with treatment, complete resolution upon withdrawal with treatment, C-D-R protocol, Naranjo algorithm, time for complete or partial resolution of the oral lesions, and follow-up. The duration of follow-up was either recorded as mentioned by the authors or calculated on the basis of the initial date the patient was in complete clinical remission and discontinued systemic therapy to the last date of observation reported in the reviewed paper.

Statistical analysis

Descriptive statistics were calculated on 54 patients with DIOLRs described in 40 case reports/series, including mean ± standard deviation for age, medians with interquartile range for timing of latency period, partial/complete resolution of oral lesions, and last follow-up, and frequency distribution for offending and concurrent medication distribution, clinical characteristics, and algorithms used for adverse drug reactions.

Mann-Whitney *U* test was used to determine the difference of medians of latency period and time partial/complete resolution of oral lesions between males (M) and females (F), whereas Spearman's correlation (ρ) was used to identify any correlation between age and latency period and between age and time partial/complete resolution of oral lesions. Conversely, Kruskal-Wallis analysis was used to evaluate the difference of medians between age, latency period, and time partial/complete resolution of oral lesions by number of concurrent mediation(s).

Lastly, a new software called Bibliometrix (http://www.bibliometrix.org/) was used to analyze the frequency of the most used words to describe the clinical and histopathological features of DIOLRs.

Results

Literature search

As shown in Fig. 1, we selected 155 full-text articles out of 810 screened. Of those 155, only 46 were included for qualitative synthesis: 40 case reports/case series, to a total of 54 patients described, and 6 studies. No cohort or case-control studies were found.

Characteristics and methodological quality of the included case reports/series and studies

The characteristics of all included case reports/series and studies are described in Tables 1, 2, and 3. Statistical analyses was done only for case reports/series, because the six studies lacked of the minimal information to run any analyses: data were confused and scanty. All cases (40 reports and 6 studies) were retrospective in nature.

In case reports/series, there was a slightly higher frequency in the females (53.7%) than males (44.4%), with a mean age of 56.81 (SD 11.72) and the highest number of cases reported in Australia (25.5%) followed by USA (19.1%) and Italy (14.9%) (Table 1).

The median number of culprit medication(s) described was 1 (Q1–Q3, 1–3) (Table 1), with the most frequent ones being Methyldopa with 11 cases (20.37%), Interferon (IFN)-alpha with 6 cases (11.11%), and Imatinib and Infliximab with 5 cases (9.26%) (Table 2).

The median number of concurrent medication(s) described was 2 (Q1–Q3, 1–3) (Table 1), with the most frequent ones being Potassium chloride with 5 cases (14.29%) and Allopurinol and vitamin C with 3 cases (8.57%) (Table 2).

Interestingly, only 8 (14.8) cases showed no concurrent medications other than the culprit, whereas in 27 other cases, 1 to 7 concurrent medications were reported (Table 1). Similarly, in only 43 (83.3%) of cases, there was one single culprit medication, while in the remaining 9 cases (16.37%), it varied from 2 to 4 (Table 1).

Additional information about the frequency of cutaneous and other mucous membrane involvement, history of personal and family allergy, and direct immunofluorescence (DIF) are also described in Table 3.

Causality assessment, latency period, and complete/partial resolution of the oral lesions

The causality assessment was done only in 8 (14.8%) cases with the C-D-R protocol. Those 8 cases turned out to be positive upon rechallenge of the alleged offending medication. This protocol was not reported in 33 cases (61.1%) and was not done in 12 (22.2%). The Naranjo algorithm was not reported in the majority of cases (98.2%). Both algorithms were not applicable in 1 case because the culprit medication was hepatitis B vaccine, and therefore, no dechallenge–rechallenge could be performed (Table 3).

The suspected offending medication was withdrawn in only 37 (68.5%) of the cases, obtaining a partial or complete resolution without treatment in only 18 (16.7%) cases and with treatment in 30 (27.7%) cases (Table 3).

The median time for latency period was 165 days (Q1–Q3, 60–406), for partial or complete resolution of the oral lesions was 56 days (Q1–Q3, 28–90), and for the follow-up was 8.5 months (Q1–Q3, 2.12) (Table 1).

The analysis of variance (ANOVA-Mann-Whitney) showed no statistical difference between male (M) and female (F) patients in terms of latency period ($p = 0.984$) but showed a difference in TPCR ($p = 0.030$) with males healing faster than females. Similarly, Spearman's correlation showed no statistical difference among patients of different age in terms

Table 1 Demographics, medications, and timing of latency and partial/complete resolution of oral lesions on 54 DIOLR case reports

Variables				Number (%)
Gender	Male			24 (44.4)
	Female			29 (53.7)
	NR			1 (1.9)
Country	Australia			12 (25.5)
	USA			9 (19.1)
	Italy			7 (14.9)
	UK			6 (12.8)
	Other countries			13 (27.7)
	Number of cases	Mean	SD	Min–Max
Age	54	56.81	11.72	15.76
		Median	Q1–Q3	Min–Max
Latency period (days)	50	165	60–406	7–1460
TPCR (days)	37	56	28–90	14–392
Last follow-up (months)	30	8.5	2–12	0.8–72
Culprit medications	54	1	1–3	1–4
Concurrent medications	34	2	1–3	0–7
				Number of cases (%)
Distribution of culprit medications	1			45 (83.3)
	2			6 (11.1)
	3			2 (3.7)
	4			1 (1.9)
Distribution of concurrent medications	NR			19 (35.2)
	0			8 (14.8)
	1			9 (16.7)
	2			8 (14.8)
	3			4 (7.4)
	4			5 (9.3)
	7			1 (1.8)

Legend: **TPCR**, Time for partial/complete resolution; **SD**, Standard deviation; **Q1–Q3,** Quartiles 1 and 3; **NR,** Not reported

of latency period ($\rho = 0.272$; $p = 0.056$) and TPCR ($\rho = -0.001$; $p = 0.993$) (Table 4).

Interestingly, the ANOVA (Kruskal-Wallis test) demonstrated that the medians of age ($p = 0.103$), latency period ($p = 0.947$), and TCCR ($p = 0.982$) do not vary significantly among patients taking none, one, or more concurrent medications (Table 5).

Clinical and histopathological features

The analysis of the frequency of the most frequent words used to described the clinical and histopathological features of DIOLRs revealed that "mucosa," "buccal," "tongue," and "white" are the most frequent ones, unlike "bilateral" and "erosive" (Table 6) (Fig. 2). Histologically, the most frequent words were "infiltrate," "lymphocytes," "cell," and "basal," whereas "band-like," "degeneration," and "layer" were the least frequent (Table 5) (Fig. 3).

Studies

Only six studies were considered eligible for this systematic review, whose data are tabulated in Tables 7 and 8. All of them were published between 1983 and 2009, two coming from UK and one each from Canada, Turkey, USA, and Scotland. The number of patients who took medications and presented with oral lesions varied from 7 to 175. Two studies [24, 28] found that the culprit medication was Penicillamin, with six concurrent medications in the first study and no concurrent medications in the second one.

In the other four studies [16, 25–27], the main culprit medication was a non-steroidal anti-inflammatory drug (NSAID) identified as Naproxen in only one study, in which 6 out of 55 patients showed oral lesions only and 22 showed both oral and cutaneous lesions, with the majority of patients taking multiple concurrent medications [26].

Table 2 Distribution of culprit and concurrent medications

Culprit medication	Frequency	Percent
Methildopa	11	20.37
Interferon-alpha	6	11.11
Imatinib	5	9.26
Infliximab	5	9.26
Chlorpropamide	3	5.56
Lithium carbonate	3	5.56
Ribavirin	3	5.56
Concurrent medication	Frequency	% on 34 cases
Potassium chloride	5	14.29
Allopurinol	3	8.57
Vitamin C	3	8.57
Chlorothiazide	2	5.71
Amitriptyline	2	5.71
Digoxin	2	5.71
Metformin	2	5.71
Nitrazepam	2	5.71
Stavudine	2	5.71

In the other three studies, the NSAID was not specified and taken only by a small portion of the entire study population: 24 out of 176 cases [25], 16.9% of cases calculated on the entire sample size 225, although only 119 took medications [16] and in 6 out of 49 cases [27] (Table 7).

Withdrawal of the alleged offending medication was only done in 3 out of 6 studies [16, 24, 28] but not in all patients: 6 out of 7 cases obtaining a complete clinical remission in all 6 [24]; 12 out of 119 cases, obtaining partial remission in 8 and complete in 4 [16]; and 8 out of 10 cases, obtaining a complete remission [28]. The C-D-R protocol was done in 3 out of 119 in one study who were positive upon rechallenge [16], whereas in another study, the majority of tested patients were positive upon rechallenge, but unclear the exact number of tested patients [28]. Naranjo algorithm was not reported in any of the 6 studies (Table 8).

A more detailed description of the characteristics of all case reports/series and studies are summarized in Tables 7 and 8.

Discussion

We found that no standardized protocol has been used in all the reviewed cases to conclusively prove this causative relationship. Indeed, the C-D-R protocol was only done in 14.8% of the reviewed case reports/series, while Naranjo algorithm was never reported. Therefore, our analyses failed to provide qualitative evidence sufficient enough to demonstrate the scientific validity of the entity called DIOLRs, as the available data turned out to be scanty, confusing, and inaccurately collected.

Similarly, the dechallenge (withdrawal) of the alleged offending medication was performed in only 68.5% of case reports/series, resulting in a partial or completed resolution of oral lesions without additional treatment in only 16.7% of case reports/series.

The quality of the six studies included was very poor: the actual number of patients with DIORLs was unclear in the majority of them, as well as the culprit medication, that was specified in just three out of six studies. In the other three

Table 3 Clinical characteristics, withdrawal, resolution of oral lesions, and algorithm used on 54 DIOLR case reports

	Yes (%)	No (%)	NR (%)	NA (%)	ND (%)
Presence of cutaneous lesions	13 (24)	11 (20)	30 (56)	–	–
Presence of other mucous membranes involved	3 (5.5)	8 (14.8)	43 (79.7)	–	–
Personal history of allergy	0 (0.0)	4 (7.4)	50 (92.6)	–	–
Family history of allergy	0 (0.0)	1 (1.8)	53 (98.2)	–	–
DIF evaluation	11 (20.3)[a]	–	25 (46.3)	–	18 (33.4)
Withdrawal	37 (68.5)	8 (14.8)	5 (9.3)	4 (7.4)[b]	–
Partial resolution without treatment	9 (16.7)	3 (5.5)	3 (5.5)	39 (72.3)	–
Complete resolution without treatment	9 (16.7)	3 (5.5)	3 (5.5)	39 (72.3)	–
Partial resolution with treatment	19 (35.1)	5 (9.3)	5 (9.3)	25 (46.3)	–
Complete resolution with treatment	11 (20.3)	5 (9.3)	5 (9.3)	33 (61.1)	–
C-D-R	8 (14.8)[c]	–	33 (61.1)	1 (1.9)	12 (22.2)
Naranjo algorithm	–	–	53 (98.2)	1 (1.8)	–

Legend: **NR**, Not reported; **NA,** Not applicable; **NR,** Not reported; **C-D-R,** Challenge–dechallenge–rechallenge; **DIF**, Direct immunofluorescence

[a] DIF analyses was done in only 11 cases, but turned out to be positive in only 5 (45.5%) out of 11.

[b] 3 out of 4 cases had their medication cycle completed, and 1 out of 4 cases had a vaccination for Hepatitis B completed. Therefore, withdrawal was not applicable in those 4 cases.

[c] C-D-R protocol was done in only 8 cases and it turned out to be positive in all of them.

Table 4 Distribution of latency period and partial/complete resolution by gender and age

	Groups gender	Median	Q1–Q3	*p* value
Latency (days)	M	180.0	60.0–365.0	0.984
	F	135.0	50.8–730.0	
TPCR (days)	M	28.0	14.0–75.0	0.030*
	F	60.0	28.0–150.0	
	Spearman's correlation (ρ) with age			
Latency (days)		0.272		0.056
TPCR (days)		− 0.001		0.993

Legend: **TPCR,** Time for partial/complete resolution; **Q1–Q3**, Quartiles 1 and 3; **M**, Males; **F,** Females

*Significant $0.01 < p$ value ≤ 0.05

Table 6 Top ten frequent words used to describe clinical and histopathological feature of DIOLRs

Clinical oral features			Histopathological features		
Rank	Word	Frequency	Rank	Word	Frequency
1°	Mucosa	55	1°	Infiltrate	40
2°	Buccal	38	2°	Lymphocyte	25
3°	Tongue	31	3°	Cell	19
4°	White	31	4°	Basal	16
5°	Ulcer	25	5°	Inflammatory	16
6°	Lesion	22	6°	Epithelium	15
7°	Plaque	15	7°	Lichenoid	12
8°	Right	14	8°	Acanthosis	11
9°	Area	12	9°	Band-like	10
9°	Bilateral	12	9°	Degeneration	10
10°	Erosive	11	10°	Layer	8

studies, just the category of the suspected offending medication and not a specific culprit drug was mentioned. Concurrent multiple medications were reported only in 2 out of 6 studies, but it was impossible to establish how many patients took how many and which concurrent medications. Latency period was only specified in three out of six studies. More importantly, only three out of six studies reported a dechallenge that was only done in some patients, as well as the C-D-R protocol, performed in only two out of six studies and not on all patients. Scanty information regarding the partial and/or complete clinical remission with and without therapy was provided, whereas the use of any algorithm to identify an ADR was completely absent as well as the follow-up time. Because of the paucity of data, no statistical analysis was performed.

Interestingly, the analysis of the most common terminology used to describe the clinical and histopathological features of DIOLRs revealed that those lesions seem to affect primarily the buccal mucosa and the tongue with more keratotic and less erosive clinical features, unlike a previous investigation [16]. Interestingly, the bilaterality component does not seem to be predominant, in line with another study [17]. Histologically, the inflammatory infiltrate seems to be predominantly lymphocytic and not mixed with plasma cells and eosinophils as previously reported [12, 18].

Our review demonstrated that the most common culprit medications are Methyldopa, IFN-alpha, Imatinib, and Infliximab, because probably they fitted more our stringent inclusion criteria than other cases. Additionally, it is likely that more cases about these abovementioned drugs have been published based on previous reports, although not all of them followed specific

Table 5 Analysis of variance of age, latency period, and partial/complete resolution by concurrent medication(s)

	Concurrent medications	Number	Median	Q1–Q3	*p* value
Age	0	7	62.0	51.0–65.0	0.103
	1	9	51.0	42.0–60.0	
	2	8	65.0	55.5–71.0	
	> 2	10	62.0	50.0–69.8	
	NR	20	58.0	53.3–61.8	
Latency period (days)	0	7	270.0	14.0–730.0	0.947
	1	8	112.5	45.0–202.5	
	2	8	270.0	35.0–652.5	
	> 2	9	150.0	67.5–1330.0	
	NR	18	150.0	59.0–456.25	
TPCR (days)	0	6	51.0	19.3–199.3	0.982
	1	6	24.5	19.3–365.0	
	2	6	50.0	24.5–90.0	
	> 2	8	45.0	28.0–90.0	
	NR	11	56.0	28.0–90.0	

Legend: **TPCR**, Time for partial/complete resolution; **Q1–Q3**, Quartiles 1 and 3

Fig. 2 Clinical oral features: word cloud of the most frequent terms

algorithms/protocols to ascertain a true causative relationship. Indeed, interestingly, only 2 out of 11 cases of Methyldopa, 1 out of 6 cases of IFN-alpha, and 1 out of 5 cases of Imatinib turned out to be positive upon rechallenge (Tables 2 and 8).

The wide discrepancy between the long list of medications considered capable of causing OLLs [3, 4, 7, 12, 13] and the very few medications indicated as culprit in our review may reside in the fact that specific diagnostic criteria for DIORLs are still lacking and not specific [12], and the over-published cases of DIORLs have been selected without following any guidelines, which should include accurate description of the oral clinical features, histopathological exam in all cases with adjunct direct and indirect immunological testing to exclude conditions, such as discoid lupus erythematosus [29].

Although DIOLRs are not life-threatening conditions, thorough information regarding the patient's past medical history, including also past personal and family history of allergies, as well as an accurate and complete drug history, including not only the suspect offending medication(s) but also the concurrent one(s), whether they are prescribed, or over-the-counter, or alternative remedies, might help in reducing the chance of developing DIOLRs. If a suspect of DIOLRs arises, we must also include the timing, i.e., the first onset of symptoms, although these might not necessarily coincide with the onset of OLLs; however, the simplistic timing from introduction or withdrawal of medication to the unpleasant reaction might not be sufficient alone to prove conclusively such causality relationship; other parameters of timing must be considered as well as the pattern of an adverse event and further investigations, such as blood test or biopsies [2].

Fig. 3 Histopathological features: word cloud of the most frequent terms

For this reason, in order to ascertain the causality assessment in any suspected drug-induced adverse reaction, the use the C-D-R protocol or one of the many internationally accepted algorithms/protocols for ADRs, such as APS [22], should be mandatory [29], although they both have drawbacks. Indeed, the C-D-R protocol in some cases can be unethical and potentially dangerous, because there might not be a valid pharmacological alternative, and there might be a risk of anaphylactic reactions. Additionally, in DIOLRs, oral reactions may take months to resolve upon discontinuation of the alleged offending medication [4, 18], making even more difficult to interpreter any possible causative relationship. On the other hand, these standardized protocols/algorithms for ADR are usually unable to produce a precise and reliable quantitative estimation of relationship likelihood, although they are still used as a common routine procedure in pharmacovigilance [21].

Despite this, of course, we cannot and must not neglect the use of one of the standardized algorithms, such as APS [22], along with specific guidelines published by the American Society of Health-System Pharmacists [30] on how to monitor and report an ADR. This would reduce the risk of publishing new cases of DIOLRs based on previous reports that did not follow any algorithm/protocol for ADRs.

We concur with previous authors [29] that the analysis of cases of can no longer be based on case reports, reports of small series, and analytical reports, where a meta-analysis cannot be performed. Ideally, we need robust and well-constructed prospective cohort study with a large sample of OLLs cases vs an age- and sex-matched cohort in order to determine whether there is a true causal relationship between drug ingestion and OLLs onset or whether this association is merely coincidental [29].

However, case reports/case series and/or retrospective studies can still add important information on DIOLRs and should not be neglected, only if well-conducted and well-documented, in order to be able to draw a definitive conclusion of a certain association between any medication and OLLs.

Limitations

Our review has several limitations, mainly related to the fact that our analysis is only based on case reports, since the included studies presented with incomplete data for an accurate analysis regarding drugs and OLLs. Because of their nature, those case reports are unable per se of depicting an accurate clinical overview of this entity. Case reports represent the lowest level of scientific evidence; they can be biased by the authors as well as reviewers who based their assumption on

Table 7 Case reports/series and studies on oral drug-induced lichenoid reactions (DIOLRs) from 1966 to 2016 included in this review

Case reports/series of DIOLRs							
References/Year	Country of publication	Age	Gender	Offending medication	Underlying medical conditions	Concurrent medications at the time of DIOLRs onset	Latency period (days)
Dinsdale et al. 1968	UK	53	M	CPM	DM II	NR	120
Burry and Kirk 1974	Australia	66	F	Methyldopa	NR	NR	Several years
		68	F	Methyldopa	NR	NR	Several years
Hay and Reade 1978	Australia	75	F	Methyldopa	HTN	CTZ, KCl	270
		65	F	Methyldopa	HTN	None	365
		71	F	Methyldopa	HTN	NZP, Spironolactone	270
		63	F	Methyldopa	HTN	Amitriptyline, Indomethacin	730
		76	M	Methyldopa	HTN	Theophylline, DGX, KCl	1460
		67	F	Methyldopa	HTN	Prindolol, Aspirin	810
		68	F	Methyldopa	HTN	None	1460
		69	F	Methyldopa	HTN	Doxepin, KCl, CTZ	1380
Wiesenfeld et al. 1982	Scotland	61	F	Oxprenolol, CPT, Methyldopa	Ferguson-Smith disease, HTN	NR	14–28
Hamburger and Potts 1983	UK	64	M	Indomethacin	RA	None	270
Baer and Goldenhar 1983	USA	61	M	Methyldopa	HTN, TIA, CVA	CTL, ALP, FLZ, KCl	1280
Ferguson et al. 1984	Scotland	60	F	Fenclofenac	Osteo-arthritis, UTI	Amoxicillin	730
Hogan et al. 1985	Canada	41	M	Lithium Carbonate	Dep, OCD, Acne	Desipramine	90–120
Colvard et al. 1986	USA	51	F	Clonazepam	Anxiety	None	730
Firth et al. 1989	Australia	63	F	Enalapril	HTN, Arthritis	ALP, CCH, QTZ, KCl	30–120
	Australia	67	M	Captopril	NR	NFD, DGX, Vit C, NZP	300
Lamey et al. 1995	Scotland	58	F	Met, Bendrofluazide, Debrisoquine	HTN, DM II	NA	Impossible to determine
	Scotland	63	F	CPM, Atenolol	HTN, DM II, Chronic bronchitis	Salbutamol	Impossible to determine
Papini et al. 1994	Italy	60	M	IFN-alpha	HCV	NR	10
Nunez et al. 1995	Spain	45	F	IFN-alpha	HCV	None	14
Nagao et al. 1996	Japan	54	F	IFN-alpha	Chronic active HCV	NR	77
	Japan	54	F	IFN-beta	Chronic active HCV	NR	56
Kutting et al. 1997	Germany	74		IFN-alpha	Mycosis fungoides	NR	240
Pemberton et al. 2000	UK	48	F	HBV vaccine	Non-contributory	Estradiol	120
Varela et al. 2000	Portugal	38	M	IFN-alpha	HCV, HBV	NR	1460
Lim and Muir 2002	Australia	72	F	Imatinib	CML	APL, Moclobemide, OPZ	150
Guijarro-guijarro et al. 2003	Spain	71	F	Clopidogrel	CI, HTN, HPL, Dep	Mianserin, Felodipine	7
Ena et al. 2004	Italy	62	M	Imatinib	GIST	NR	195

Table 7 (continued)

Case reports/series of DIOLRs							
References/Year	Country of publication	Age	Gender	Offending medication	Underlying medical conditions	Concurrent medications at the time of DIOLRs onset	Latency period (days)
Armour and Lowe 2005	Australia	52	M	IFN-alpha, G-CSF, Ribavirin, Propranolol	HCV	None	70–140
Campisi et al. 2005	Italy	51	M	Lithium carbonate	Bipolar disorder	Oxcarbazepine	210
Moss et al. 2007	USA	55	F	Infliximab	Crohn's disease	AZA, Mesalamine	35
De Simone et al. 2008	Italy	62	M	Adalimumab	Psoriasis, PA	NR	28
Oiha et al. 2008	USA	51	M	IFN-alpha	HCV, HTN, Dep, Melanoma	Ribavirin, Amlodipine, Sertraline	NR
Giuliani et al. 2008	Italy	50	M	IFN-alpha, Ribavirin	HIV, HCV	Stavudine, Didanosine	420
	Italy	47	M	IFN alpha, Ribavirin	HIV, HCV	Stavudine, Lamivudine, Zidovudine, Nevirapine	90
Woo et al. 2009	USA	15	M	Carbamazepine, Risperdone	Sleep and mood disorder	NR	365
Basso et al. 2009	Brazil	55	M	Imatinib	CML	NR	90
Asarch et al. 2009	USA	55	M	Infliximab	PA, Psoriasis, HTN, HCL, ED, CBP	MTP, ATV, HDC, Sulindac, Amitriptyline, CPDL, SDF	60
Kaomongkolgit 2010	Thailand	58	M	Atenolol, CPM	HTN, DM II	NR	180
Artico et al. 2011	Brazil	39	M	Carbamazepine	AWS	Fluoxetine	180
Mocciaro et al.2011	Italy	59	M	Certolizumab	Crohn's disease	NR	730
Brazzelli et al. 2012	Italy	60	F	Imatinib	CML	Prednisone	90
Bugueno et al. 2013	USA	25	F	Lithium carbonate	BD, ADHD, Hypersomnia, SCT	Bupropion, Lamotrigine, Vit C, Omega-3 fatty acid	28
Kuten-Shorrer et al. 2014	USA	43	F	Rituximab	Follicular NHL	Vit C	28
Polat and Ustun 2014	Turkey	46	F	Imatinib	CML	NR	90
Hammami et al. 2015	Tunisia	52	M	Glimepiride	DM II	Metformin	30
Kadam et al. 2015	India	57	M	Duloxetine	HTN, DM II, Dep	Atenolol, Met	few weeks
Andrade et al. 2015	Portugal	60	F	Infliximab	Crohn's disease	NR	NR
	Portugal	52	F	Adalimumab, Infliximab	Crohn's disease	NR	180
	Portugal	54	F	Infliximab	Ulcerative colitis	NR	60
Thompson et al. 2016	USA	62	M	Secukinumab	Psoriasis	None	7

Table 7 (continued)

Studies on DIOLRs								
References	Country of publication	Number of patients	Age	Gender	Offending medication	Underlying medical conditions	Concurrent medications at the time of DIOLRs onset	Latency period (days)
Blasberg et al. [24]	Canada	7	Range, 32–68 Mean, 55.5 SD, 12.5	6 F 1 M	Penicillamine	Rheumatoid arthritis	ASA, KET, NAP, INM, IBU, PDN	90–365
Clayton et al. [25]	UK	175	Range, 24–86 Mean, 56 calculated only on 106	All F	NSAIDs (24/175)	NR	NR	NR
Gunes et al [26]	Turkey	55 (6 with oral lesions only 22 with oral and skin lesions)	Range, 8–75 Mean,48.8 calculated on all 56	32 F 23 M	Naproxen	NR	12/55 patients took (ASA, INM, IBU, KTZ, ENL, FUR; 8/55 patients took MAX, AMX-CL, STC, AZM, PDP, CEF, PCTM, CLA, TZN, OXLM, CHZX, MPX, CPHN, DPR, MTZ, FAM, NIM, ETD, ALN, DCFN,.)	14–90 calculated on all patients
Potts et al. [16]	UK	225 (only 119 took meds)	Mean, 54.9 SD, 13.6 calculated on 225 pts	153 F 72 M	NSAIDs (16.9% calculated on all 225 pts)	NR	NR	NR
Robertson and Wray [27]	Scotland	49	Mean, 51.9 SD, 13.7	29 F 20 M	NSAIDs, 6 pts Anti-HTN, 10 pts Psychotropic drugs, 6 pts Other meds, 21 pts	Rheumatoid arthritis and non-rheumatoid arthritis	Several patients took more than 1 drug	NR
Powell et al. [28]	USA	10	NR	NR	Penicillamine	Primary biliary chirrosis	None	60–1095

Legend: **ADHD,** attention deficit/ hyperactivity disorder; **ALN,** Alendronate; **ALP,** Allopurinol; **AMX,** Amoxicillin; **AMX-CL,** Amoxicillin and clavulanate; **ASA,** Acetylsalicylic acid; **ATV,** Atorvastatin; **AWS,** Alcohol withdrawal syndrome; **AZA,** Azathioprine; **AZM,** Azithromycin; **BAN,** Bendrofluazide; **BD,** Bipolar disorder; **CBP,** Chronic back pain; **CCH,** Colchicine; **CEF,** Cefaclor; **CHZX,** Chlorzoxazone; **CI,** Cerebral infarction; **CLA,** Clarithromycin; **CML,** Chronic myeloid leukaemia; **CPDL,** Carisoprodol; **CPHN,** Chlorpheniramine; **CPM,** Chlorpropamide; **CPT,** Cyclopenthiazide; **CTL,** Chlorthalidone; **CTZ,** Chlorothiazide; **CVA,** Cerebrovascular accident; **DCFN,** Diclofenac; **Dep,** Depression; **DGX,** Digoxin; **DM II,** Type II Diabetes; **DPR,** Dipyrone; **ED,** Erectile dysfunction; **ENL,** Enalapril; **ETD,** Etodolac; **FAM,** Famotidine; **FLZ,** Flurazepam; **FUR,** Furosemide; **G-CSF,** Granulocyte-colony stimulating factor; **GIST,** Gastrointestinal Stromal tumor; **HBV,** Hepatitis B Virus **HCL,** Hypercholesterolemia; **HCV,** Hepatitis C Virus; **HDC,** Hydrocodone; **HIV,** Human Immunodeficiency Virus; **HPL,** Hyperlipidemia; **HTN,** Hypertension; **IBU,** Ibruprofen; **IFN**, Interferon; **INM,** Indomethacin; **KCl,** Potassium chloride; **KET,** Ketoprofen; **KTZ,** Ketoconazole; **Met,** Metformin; **MPX,** Mephenoxalone; **MTP,** Metoprolol; **MTZ,** Metronidazole; **NAP,** Naproxen; **NFD,** Nifedipine; **NHL,** Non-Hodgkin Lymphoma; **NIM,** Nimesulide; **NZP,** Nitrazepam; **OCD,** Obsessive compulsive disorder; **OPZ,** Omeprazole; **OXLM,** Oxolamine; **PA,** Psoriatic arthritis; **PCTM,** Paracetamol; **PDN,** Prednisone; **PDP,** Perindopril; **QTZ,** Quinethazone; **SCT,** Sickle cell trait; **SDF,** Sildenafil; **STC,** Sultamicillin; **TIA,** Transient Ischemic Attack; **TZN,** Tizanidine; **UTI,** Urinary tract infection; **Vit C,** Vitamin C

Table 8 Withdrawal of the offending medication, treatment, remission, and adverse drug-reaction algorithms

Case reports/series of DIOLRs										
References	Withdrawal	PCR without treatment	CCR without treatment	Treatment	PCR with treatment	CCR with treatment	C-D-R	Naranjo	TP/CCR (days)	Last FU (months)
Dinsdale et al.	Yes	NA	Yes	NR	NA	NA	Yes (+)[a]	NR	14	1
Burry and Kirk	Yes	NA	NA	PLN	Yes	NA	NR	NR	NR	4
	Yes	NA	NA	PLN	Yes	NA	NR	NR	NR	5
Hay and Reade	Yes	Yes	NA	HCS mouthwash, TAI, Amph B, Nystatin oint	No	No	NR	NR	150	12
	Yes	Yes	NA	TAI, HCS topical	Yes	NA	Yes (+)[a]	NR	60	NR
	Yes	Yes	NA	HCS mouthwash, TAI, Amph B, Nystatin oint	NA	No	NR	NR	60	24
	Yes	Yes	NA	Topical antifungal, Topical HCS	No	No	NR	NR	70	NR
	Yes	Yes	NA	ND	NA	NA	NR	NR	90	NR
	NR	NR	NR	NR	NR	NR	NR	NR	60	NR
	NR	NR	NR	NR	NR	NR	NR	NR	135	NR
	NR	NR	NR	NR	NR	NR	NR	NR	150	NR
Wiesenfeld et al.	Yes	NA	Yes	NR	NA	NA	NR	NR	180	6
Hamburger and Potts	Yes	NA	NA	CHX, Ibuprofen, PLN mouthwash	NA	Yes	Yes (+)[a]	NR	14	NR
Baer and Goldenhar	Yes	Yes	NA	ND	NA	NA	NR	NR	28	NR
Ferguson et al.	Yes	NA	Yes	ND	NA	NA	ND	NR	28	12
Hogan et al.	Yes	NA	Yes	ND	NA	NA	Yes (+)[a]	NR	21	12
Colvard et al.	No	No	No	ND	NA	NA	ND	NR	NA	NR
Firth et al.	Yes	NA	NA	Antifungal therapy	Yes	NA	NR	NR	90	3
	Yes	Yes	NA	NR	NA	NA	NR	NR	30	1
Lamey et al.	NR	No	No	BET systemic	No	No	NR	NR	NA	NR
	No	NA	NA	NR	NR	NR	ND	NR	NR	NR
Papini et al.	Yes	NA	NA	Systemic CS and CsA	Yes	NA	NR	NR	NR	NR
Nunez et al.	Yes	NA	Yes	NR	NA	NA	Yes (+)[a]	NR	21	NR
Nagao et al.	No	NA	NA	Topical CS	Yes	NA	ND	NR	NR	12
	Cycle completed	NA	NA	NR	NR	NR	NR	NR	NR	12
Kutting et al.	Yes	NA	Yes	ND	NA	NA	NR	NR	56	2
Pemberton et al.	NA	NA	NA	Topical BET	NA	Yes	NA	NA	365	12
Varela et al.	Yes	NA	Yes	NR	NA	NA	NR	NR	NR	12
Lim and Muir	Yes	NA	NA	Topical DXT	Yes	NA	ND	NR	28	1.5
	Yes	NA	NA	Topical TA, DFZ	Yes	NA	NR	NR	NR	NR

Table 8 (continued)

Case reports/series of DIOLRs										
References	Withdrawal	PCR without treatment	CCR without treatment	Treatment	PCR with treatment	CCR with treatment	C-D-R	Naranjo	TP/CCR (days)	Last FU (months)
Guijarro-guijarro et al.										
Ena et al.	No	NA	NA	Topical CS, OPZ	NA	Yes	ND	NR	280	10
Armour and Lowe	Yes (all but propranolol)	NA	NA	PDN, Topical TCM	NA	Yes	NR	NR	392	14
Campisi et al.	No	No	No	Topical CBT, PDN	Yes	NA	ND	NR	NR	NR
Moss et al.	No	NA	NA	Topical TCM	Yes	NA	ND	NR	28	NR
De Simone et al.	Yes	NA	Yes	ND	NA	NA	Yes (+)[a]	NR	28	NR
Oiha et al.	Cycle completed	NA	NA	DXT elixir	NA	Yes	NR	NR	60	2
Giuliani et al.	Yes	NA	NA	CBT oint	Yes	NA	NR	NR	NR	NR
	Yes	NA	NA	CBT oint, PDN	Yes	NA	NR	NR	NR	0.8
Woo et al.	Yes	Yes	NA	DXT suspension	No	No	ND	NR	30	12
Basso et al.	No	NA	NA	Topical BCL, systemic CS	Yes	NA	ND	NR	90	8
Asarch et al.	Yes	NA	NA	CsA, PDN, Topical TA	Yes	NA	ND	NR	NR	9
Kaomongkolgit	Yes	NA	NA	Topical Flucinonide	Yes	NA	NR	NR	14	3
Artico et al.	Yes	NA	NA	Topical BCT	Yes	NA	NR	NR	21	10
Mocciaro et al.	Yes	NA	NA	Topical CS	NA	Yes	NR	NR	NR	NR
Brazzelli et al.	Yes	NA	NA	PDN, Topical CS	NA	Yes	Yes (+)[a]	NR	NR	2
Bugueno et al.	Yes	Yes	NA	ND	NA	NA	NR	NR	28	1
Kuten-Shorrer et al.	Cycle completed	NA	NA	Topical DXT, PDN, TAI, Topical CBT	NA	Yes	NR	NR	365	72
Polat and Ustun	NR	NA	NA	TAI, oral antihistaminic	Yes	NA	NR	NR	NR	NR
Hammami et al.	Yes	NA	NA	Systemic CS	Yes	NA	Yes (+)[a]	NR	14	2
Kadam et al.	Yes	NA	Yes	ND	NA	NA	NR	NR	14	15
Andrade et al.	Yes	NA	NA	PLN and Topical TA	NA	Yes	NR	NR	60	NR
	Yes	NA	NA	PLN and Topical TA	NA	Yes	NR	NR	60	NR
	No	NA	NA	Topical Fluticasone	NA	Yes	ND	NR	30	NR
Thompson et al.	Yes	NA	NA	Topical TA	Yes	NA	NR	NR	42	NR
Studies on DIOLRs										
References	Withdrawal	PCR without treatment	CCR without treatment	Treatment	PCR with treatment	CCR with treatment	C-D-R	Naranjo	TP/CCR (days)	Last FU (months)
Blasberg et al.	Yes (6 out of 7)	NA	Yes (6 out of 7)	BET oint., TA cream	Yes	NA	NR	NR	NR	NR
Clayton et al.	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR
Gunes et al.	Yes	NR	NR	NR	NR	NR	NR	NR	NR	NR
Potts et al.	Yes (12 pts)	Yes (8 pts)	Yes (4 pts)	NR	NR	NR	Yes (+)[a] (3 pts)	NR	NR	NR

Table 8 (continued)

	Case reports/series of DIOLRs									
References	Withdrawal	PCR without treatment	CCR without treatment	Treatment	PCR with treatment	CCR with treatment	C-D-R	Naranjo	TP/CCR (days)	Last FU (months)
Robertson and Wray	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR
Powell et al.	Yes (8 out of 10)	NA	Yes	NR	NR	NR	Yes (+)[a] (most patients)	NR	Several months	NR

Legend: **PCR,** Partial clinical remission; **CCR,** Complete clinical remission; **C-D-R,** Challenge-dechallgenge-rechallenge; **TP/CCR,** Time for partial/complete clinical remission; **FU,** Follow-up; **Amph B,** Amphotericin B; **BCL,** Beclomethasone; **BET,** Betamethasone; **CBT,** Clobetasol; **CHX,** Chlorxedine; **CS,** Corticosteroid; **CsA,** Cyclosporine; **DFZ,** Deflazacort; **DXT,** Dexamethasone; **HCS,** Hydrocortisone; **Oint,** ointment; **OPZ,** Omeprazole; **PDN,** Prednisone; **PLN,** Prednisolone; **TA,** Triamcinolone; **TAI,** Triamcinolone injections; **TCM,** Tacrolimus; **ND,** Not done; **NA,** Not Applicable; **NR,** Not reported.

[a] This indicate that a C-D-R protocol has been used, giving a positive response upon rechallenge of the allegedly offending medication

previous similar reports, which might not have used any of the many standardized protocol/algorithm to assess an ADR.

Another important limitation is that those included cases were not homogenous: not all cases described information regarding all the variables tabulated in Tables 7 and 8.

Additionally, many possible cases of DIOLRs might or might not have been reported only because previous reports on the same medication had already been described in the literature.

We excluded all those muco-cutaneous cases where an oral biopsy was not performed. This might have introduced a bias in reporting other cases of DIOLRs; however, it would have been impossible to truly ascertain the nature of the oral lesions, without a histopathological confirmation from an oral biopsy. If this had been performed in all muco-cutaneous cases, we would have probably had a different number of cases to be included in our analysis.

Lastly, we still do not have standardized and validated diagnostic criteria for DIOLRs; there is no consensus statement on this entity. Therefore, if such criteria had existed, some of those case reports would have not fit those criteria, while some others would, thus making the total number of cases included in this review overrepresented or underrepresented.

Conclusions

Over the past decades, we have very often based our causality assessment of DIOLRs on previous cases, assuming that they were reported following a scientific methodology, but many of them clearly were not, or based on cases published in the dermatological literature, where oral manifestations might not always be present and/or an oral biopsy was not always performed.

Rebus sic stantibus, we cannot claim that DIORLs—as a new and separate nosological entity—does exist or that does not, because of the lack of a scientific methodology applied by clinicians over the past 50 years in diagnosing DIOLRs. This has created an enormous confusion and provided scarce and inaccurate information, as it seems that the majority of those reports have claimed an association just anecdotally, and it is now no longer time for anecdotes, but for science.

Therefore, we should no longer talk about DIORLs until we will be able to prove its existence conclusively.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

References

1. WHO (1972) International drug monitoring: the role of national centres. Rep Ser 498
2. Edwards IR, Aronson JK (2000) Adverse drug reactions: definitions, diagnosis, and management. Lancet 356(9237):1255–1259. https://doi.org/10.1016/S0140-6736(00)02799-9
3. Zelickson BD, Rogers RS 3rd (1986) Drug reactions involving the mouth. Clin Dermatol 4(1):98–109
4. Boras VV, Andabak-Rogulj A, Brailo V, Simunkovic SK, Gabric D, Vrdoljak DV (2015) Adverse drug reactions in the oral cavity. Acta Clin Croat 54(2):208–215
5. Yuan A, Woo SB (2015) Adverse drug events in the oral cavity. Oral Surg Oral Med Oral Pathol Oral Radiol 119(1):35–47. https://doi.org/10.1016/j.oooo.2014.09.009
6. Femiano F, Lanza A, Buonaiuto C, Gombos F, Rullo R, Festa V, Cirillo N (2008) Oral manifestations of adverse drug reactions: guidelines. J Eur Acad Dermatol Venereol 22(6):681–691. https://doi.org/10.1111/j.1468-3083.2008.02637.x
7. Abdollahi M, Rahimi R, Radfar M (2008) Current opinion on drug-induced oral reactions: a comprehensive review. J Contemp Dent Pract 9(3):1–15
8. Belvederi Murri M, Guaglianone A, Bugliani M, Calcagno P, Respino M, Serafini G, Innamorati M, Pompili M, Amore M (2015) Second-generation antipsychotics and neuroleptic malignant syndrome: systematic review and case report analysis. Drugs R D 15(1):45–62. https://doi.org/10.1007/s40268-014-0078-0
9. Healy CM, Smyth Y, Flint SR (2004) Persistent nicorandil induced oral ulceration. Heart 90(7):e38. https://doi.org/10.1136/hrt.2003.031831
10. Parkhill AL (2013) Oral mucositis and stomatitis associated with conventional and targeted anticancer therapy. Aust J Pharm 1:112. https://doi.org/10.4172/2329-6887.1000112
11. Almeyda J, Levantine A (1971) Drug reactions. XVI Lichenoid drug eruptions. Br J Dermatol 85(6):604–607
12. McCartan BE, McCreary CE (1997) Oral lichenoid drug eruptions. Oral Dis 3(2):58–63
13. Serrano-Sánchez P, Bagán JV, Jiménez-Soriano SG (2010) Drug-induced oral lichenoid reactions. A literature review. J Clin Exp Dent 2:e71–e75
14. Al-Hashimi I, Schifter M, Lockhart PB, Wray D, Brennan M, Migliorati CA, Axell T, Bruce AJ, Carpenter W, Eisenberg E, Epstein JB, Holmstrup P, Jontell M, Lozada-Nur F, Nair R, Silverman B, Thongprasom K, Thornhill M, Warnakulasuriya S, van der Waal I (2007) Oral lichen planus and oral lichenoid lesions: diagnostic and therapeutic considerations. Oral Surg Oral Med Oral Pathol Oral Radiol Endod 103(Suppl 1):S25 e1–s25 e12. https://doi.org/10.1016/j.tripleo.2006.11.001
15. van der Waal I (2009) Oral lichen planus and oral lichenoid lesions; a critical appraisal with emphasis on the diagnostic aspects. Med Oral Patol Oral Cir Bucal 14(7):E310–E314
16. Potts AJ, Hamburger J, Scully C (1987) The medication of patients with oral lichen planus and the association of nonsteroidal anti-inflammatory drugs with erosive lesions. Oral Surg Oral Med Oral Pathol 64(5):541–543
17. Lamey PJ, McCartan BE, MacDonald DG, MacKie RM (1995) Basal cell cytoplasmic autoantibodies in oral lichenoid reactions. Oral Surg Oral Med Oral Pathol Oral Radiol Endod 79(1):44–49
18. Baris E, Senguven B, Tuzuner T, Gultekin SE (2014) Oral lichenoid lesions related to drugs: review of clinicopathological features and differential diagnosis. Eur J Inflamm 12(2):217–225
19. Van den Haute V, Antoine JL, Lachapelle JM (1989) Histopathological discriminant criteria between lichenoid drug eruption and idiopathic lichen planus: retrospective study on selected samples. Dermatologica 179(1):10–13
20. Naranjo CA, Busto U, Sellers EM, Sandor P, Ruiz I, Roberts EA, Janecek E, Domecq C, Greenblatt DJ (1981) A method for estimating the probability of adverse drug reactions. Clin Pharmacol Ther 30(2):239–245
21. WHO (2014) World Health Organization. JAPST (1):*22*–35 http://www.who.int/medicines/areas/quality_safety/safety_efficacy/WHOcausality_assessment.pdf (Accessed 22 June 2017)
22. Agbabiaka TB, Savovic J, Ernst E (2008) Methods for causality assessment of adverse drug reactions: a systematic review. Drug Saf 31(1):21–37
23. Moher D, Liberati A, Tetzlaff J, Altman DG, Group P (2009) Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. PLoS Med 6(7):e1000097. https://doi.org/10.1371/journal.pmed.1000097
24. Blasberg B, Dorey JL, Stein HB, Chalmers A, Conklin RJ (1984) Lichenoid lesions of the oral mucosa in rheumatoid arthritis patients treated with penicillamine. J Rheumatol 11(3):348–351
25. Clayton R, Chaudhry S, Ali I, Cooper S, Hodgson T, Wojnarowska F (2010) Mucosal (oral and vulval) lichen planus in women: are angiotensin-converting enzyme inhibitors protective, and beta-blockers and non-steroidal anti-inflammatory drugs associated with the condition? Clin Exp Dermatol 35(4):384–387. https://doi.org/10.1111/j.1365-2230.2009.03581.x
26. Gunes AT, Fetil E, Ilknur T, Birgin B, Ozkan S (2006) Naproxen-induced lichen planus: report of 55 cases. Int J Dermatol 45(6):709–712. https://doi.org/10.1111/j.1365-4632.2006.02670.x
27. Robertson WD, Wray D (1992) Ingestion of medication among patients with oral keratoses including lichen planus. Oral Surg Oral Med Oral Pathol 74(2):183–185
28. Powell FC, Rogers RS 3rd, Dickson ER (1983) Primary biliary cirrhosis and lichen planus. J Am Acad Dermatol 9(4):540–545
29. McCartan BE, McCreary CE, Healy CM (2003) Studies of drug-induced lichenoid reactions: criteria for case selection. Oral Dis 9(4):163–164
30. American Society of Health-System Pharmacists (1995) ASHP guidelines on adverse drug reaction monitoring and reporting. Am J Health Syst Pharm 52(4):417–419